A Coupled Radiative–Microphysical Model of Titan's Organic Haze: Energy Balance, a Fractal-Aggregate Scaling Law, and Their Feedback

The Titan Haze Feedback Project

Working note — compiled June 9, 2026

Abstract

Titan's organic haze is radiatively and microphysically active: it absorbs sunlight and warms the stratosphere, emits in the thermal infrared, and its vertical and size structure is set by coagulation and sedimentation of fractal aggregates whose rates depend on the local temperature. This makes the haze a genuine feedback element rather than a passive tracer. We outline a four-stage program to isolate this feedback in a one-dimensional column: (i) a correlated-k radiative energy balance for the temperature profile (a mature TAM-lineage engine, cross-validated source-by-source against an independent multi-stream DISORT/`pydisort` implementation); (ii) an analytic *scaling law* for the aerosol microphysics that maps temperature, monomer radius, and fractal dimension onto the haze number- and size-profile; (iii) iteration of (i) and (ii) to a self-consistent fixed point to diagnose the feedback; and (iv) coupling of photochemical haze production into the loop. Our central methodological result is the scaling law. Adopting a monodisperse, self-similar closure $C(z, N) = n(z)\,\delta(N - \bar{N}(z))$ for the concentration of aggregates of N monomers at altitude z, the population balance reduces from an integro-differential equation in (z, N) to a compact boundary-value problem in z for the number density $n(z)$ and mean size $\bar{N}(z)$. In the sedimentation-dominated limit it collapses further to a single first-order ordinary differential equation, $\mathrm{d}\bar{N}/\mathrm{d}z = -\beta(\bar{N}, z)\,P/[2\,\omega(\bar{N}, z)^2]$, whose asymptotic solutions grow as $\bar{N} \propto (\text{depth})^2$ in the free-molecular upper haze and $\bar{N} \propto (\text{depth})^{1/2}$ in the continuum lower atmosphere. Solving the full eddy-diffusion problem reproduces the haze extinction scale height of Tomasko et al. (2008) (64 vs. 65 km) and a sub-micron characteristic radius consistent with de Trenquelléon et al. (2025). This law is inexpensive enough to embed inside the radiative iteration, closing the feedback loop. Doing so reveals the central physical result: the coupled one-dimensional system is *bistable*. The strongly-absorbing haze is smoothly mobile in altitude with temperature (warmer stratosphere ⇒ higher haze ⇒ more absorption aloft), and this positive feedback gives the radiative–convective system two stable equilibria—a warm (~ 183 K) and a cool (~ 131 K) stratopause—between which naive iteration oscillates. We show the bistability is radiative, not microphysical (the temperature-to-haze map is smooth, while the radiative transfer is genuinely multi-valued for a fixed haze, confirmed in two independent engines), but *not robust*: a polydisperse (bimodal, size-resolved) haze, whose broad aggregate distribution sheds mass faster and thins the haze, is monostable at every distribution width sampled. The two equilibria are thus an artifact of the single-size idealization—a caution against inferring strong feedbacks from monodisperse microphysics.

1 Introduction

On Titan the photochemical haze is not a passive optical load. It absorbs solar radiation—heating the stratosphere—and emits in the thermal infrared, where it contributes a non-negligible fraction of the planetary emission and warms the troposphere (Tomasko et al., 2008).

At the same time its vertical extent and particle sizes are governed by aerosol microphysics: monomers produced high in the atmosphere coagulate into fractal aggregates that sediment downward (Cabane et al., 1993; Rannou et al., 2004; Burgalat and Rannou, 2017). Crucially, the microphysical rates are temperature dependent—the Brownian coagulation kernel scales as T/η, the gas viscosity $\eta(T)$ and mean free path $\lambda(T,P)$ enter the settling velocity, and the free-molecular kernel carries a $T^{1/2}$ factor. Temperature sets the microphysics, the microphysics sets the opacity, and the opacity sets the temperature: a closed feedback loop.

Most existing models break this loop in one direction. Radiative studies prescribe the haze from observations (Tomasko et al., 2008; Lombardo and Lora, 2023), while fully coupled treatments embed microphysics in a three-dimensional general circulation model (de Batz de Trenquelléon et al., 2025), which is faithful but expensive and difficult to interrogate. Our aim is intermediate: to isolate the one-dimensional radiative↔microphysical feedback with a fast, analytic microphysics so the coupled system can be iterated cheaply to a self-consistent state and the feedback gain measured directly.

2 Model overview

The program proceeds in four stages.

Step 1 — Radiative energy balance. A 1-D plane-parallel Titan column iterated to radiative(–convective) equilibrium for $T(z)$: shortwave absorption by CH_4 and haze; longwave cooling by gas lines, collision-induced absorption, and haze emission (Sec. 3). The production engine is a mature TAM-lineage correlated-k model (`example_bowen_fort`); we additionally built an independent multi-stream discrete-ordinates implementation (`pydisort`/DISORT; Stamnes et al., 1988) and matched the two source-by-source, so the engine is cross-validated rather than taken on trust (Sec. 3.2).

Step 2 — Microphysical scaling law. A closed relation for the steady aggregate population, coagulation + sedimentation → density/size profile, in the form $C(z,N)\,\mathrm{d}z\,\mathrm{d}N = f(z,N,T,d,D)$ (Sec. 4). This is the methodological core of the present note.

Step 3 — Coupled iteration. Map the microphysical moments to layer optical properties, feed the RT engine, update $T(z)$, and iterate; diagnose the feedback. The coupled system proves *bistable* (Sec. 5).

Step 4 — Photochemistry. Promote the imposed haze production rate to an output of a photochemical module, closing the radiation↔chemistry↔microphysics loop (Sec. 6).

The Step 1 engine is the reference correlated-k model; our DISORT implementation serves as an independent multi-stream cross-check. The two are matched to within a few percent in every opacity source on a shared 100-layer grid (Sec. 3.2), so the multi-stream solver both validates the engine and provides a higher-angular-resolution option for the scattering-dominated shortwave.

3 Radiative energy balance (Step 1)

The column spans the surface to $\sim 540\,\mathrm{km}$. Following the spectral split of Lombardo and Lora (2023), shortwave (2000–$40000\,\mathrm{cm}^{-1}$, $< 5\,\mu\mathrm{m}$) radiative transfer treats solar absorption and multiple scattering by haze (single-scattering albedo ω_0, asymmetry g), while the longwave band (1–$2000\,\mathrm{cm}^{-1}$) handles thermal cooling with gas line opacity, N_2–N_2, N_2–CH_4, CH_4–CH_4, and N_2–H_2 collision-induced absorption, and a (near-)purely-absorbing haze. Gas opacities use the correlated-k method with HITRAN line data; methane shortwave coefficients follow the

Huygens/DISR measurements supplemented above 1600 nm by laboratory data. The temperature profile is relaxed until shortwave heating balances longwave cooling.

The validation target is the energy budget of Tomasko et al. (2008): a surface net solar flux of $0.574\,\mathrm{W\,m^{-2}}$ (about 11% of the incident flux) rising to $\sim 4.1\,\mathrm{W\,m^{-2}}$ near 400 km; a net heating that exceeds cooling by at most ~ 0.5 K per Titan day near 120 km; and a thermal cooling rate peaking near 21 K per Titan day around 340 km. Roughly 80% of the top-of-atmosphere sunlight is absorbed in the column, $\sim 40\%$ below 80 km, the haze raising the planetary thermal emission by $\sim 15\%$ and reducing the tropospheric cooling rate by $\sim 30\%$ below 50 km. Baseline atmospheric and compositional inputs are collected in Table 2.

3.1 Implementation and first results

The solver is built on `pydisort` (a modern `torch`-based DISORT binding) with eight streams: a collimated solar beam in the shortwave and a band-resolved Planck thermal source in the longwave. Per layer the optical properties combine the haze and the gas. The haze extinction follows from the Step 2 microphysics through *mean-field (Rayleigh–Debye–Gans) aggregate optics*: each monomer absorbs in the Rayleigh regime, so the aggregate absorption is additive over monomers, $C_{\mathrm{abs}} = N\,C_{\mathrm{abs}}^{\mathrm{mono}}(\lambda)$ with $C_{\mathrm{abs}}^{\mathrm{mono}} = (8\pi^2 d^3/\lambda)\,\mathrm{Im}\{(m^2-1)/(m^2+2)\}$ and $m(\lambda)$ the tholin index (Khare et al. 1984; de Batz de Trenquelléon et al., 2025). This scales the extinction with monomer number (i.e. mass), not with r_a^2, and so corrects the mobility-radius cross-section that overestimated the optical depth roughly eightfold; with the monomer absorption anchored to the observed haze opacity, the visible column optical depth is ≈ 8, matching Huygens/DISR. Spectral single-scattering albedo $\omega_0(\lambda)$ and asymmetry $g(\lambda)$ come from the prescribed haze (Doose 2016): $\omega_0 \to 0$ in the UV (pure absorber), ~ 0.9 in the near-IR, near-pure absorber in the thermal IR.

The longwave combines two sources on a common 40-band IR grid (0–2000 $\mathrm{cm^{-1}}$, 50 $\mathrm{cm^{-1}}$ bands). First, *real collision-induced absorption*—the dominant thermal-IR continuum on Titan—from the HITRAN CIA coefficients (Karman et al., 2019) for N_2–N_2, N_2–CH_4, CH_4–CH_4, and N_2–H_2, as $\tau(\nu) = \sum_{\mathrm{pairs}} k_{AB}(\nu, T)\, n_A n_B\, \Delta z$; the N_2–N_2 roto-translational band makes the far-IR strongly opaque (column $\tau \gtrsim 100$ near 50–100 $\mathrm{cm^{-1}}$), the origin of Titan's pressure-induced greenhouse. Second, *correlated-k gas lines* for the stratospheric coolants CH_4, C_2H_2, C_2H_6, C_2H_4, and HCN (the same IR k-tables and 10-Gauss grid as the shortwave), summed under the perfectly-correlated assumption with each gas's VMR profile. Both gases and CIA, plus a pure-absorber IR haze, are solved per (band, Gauss-point) with a band-integrated Planck source.

The *shortwave* methane opacity is treated with correlated-k, using the same visible CH_4 k-table as the reference Fortran model (42 bands $\times$ 10 Gauss points, on a 20×10 pressure–temperature grid). Per layer the gas optical depth is $\tau = k(P,T)\,u$ with u the CH_4 column abundance in km-amagat; all 420 (band, Gauss-point) sub-intervals are solved in a single multi-wave DISORT call and recombined by the Gauss weights, with the per-band solar flux integrated from the Houghton spectrum and scaled to Titan's orbit (9.5 AU, $\sum$ bands $= 15.1\,\mathrm{W\,m^{-2}}$). The CH_4 windows let sunlight through to the surface while the bands absorb aloft; CH_4 alone absorbs $\approx 1.4\,\mathrm{W\,m^{-2}}$, peaking at ~ 14 K per Titan day in the stratosphere.

We relax the *layer-mean* temperature toward radiative–convective equilibrium: each iteration runs both bands, nudges the layer temperatures along the net heating rate (under-relaxed; the layer-mean state is in 1:1 correspondence with the per-layer heating, avoiding the level$\leftrightarrow$layer aliasing that otherwise seeds a grid-scale zig-zag), then applies a convective adjustment to a critical lapse rate ($\sim 1\,\mathrm{K\,km^{-1}}$). Cold space is imposed as the top boundary (a thermal emission temperature of 2.7 K, not the atmosphere's top temperature); without this the column spuriously absorbs downwelling longwave and overheats.

The solver reaches a genuine steady state with energy closed to ~ 0.1–$1\,\mathrm{W\,m^{-2}}$. Driven

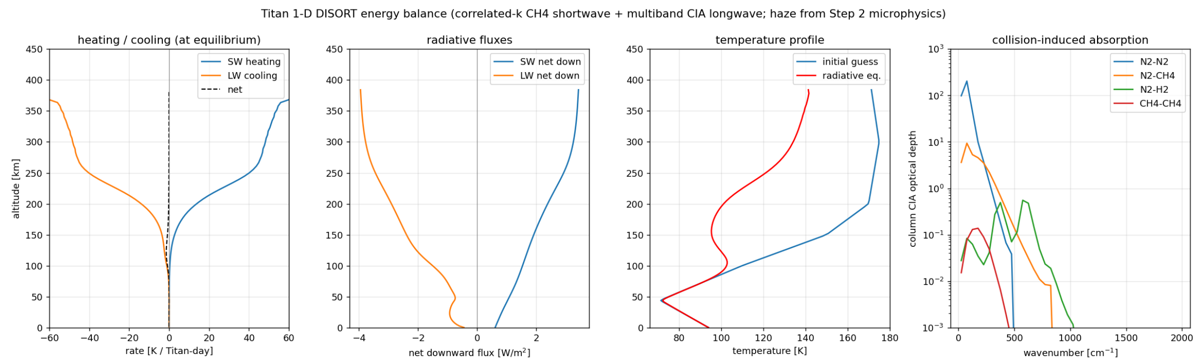

Figure 1: DISORT energy balance on the Titan column with the Step 2 haze and HITRAN collision-induced absorption. From left: shortwave heating, longwave cooling, and net rate at the relaxed state; net downward shortwave and longwave fluxes; initial-guess vs. radiative-equilibrium temperature (surface fixed); and the column CIA optical depth per pair, showing the N_2–N_2 far-IR continuum that dominates the thermal opacity.

by the Step 2 microphysics haze (through the RDG cross-section) it has the right vertical structure—a stratopause, a cold tropopause near 74 K, a ~ 92 K surface—with a stratopause near 140 K, up from ~ 115 K with the old gray cross-section but still below the observed ~ 190 K (Fig. 1, red).

To localize the residual deficit we ran the *same* radiative engine on the observationally-prescribed haze (the optical depths the reference model uses) instead of the microphysics haze. It then reaches a ~ 200 K stratopause, matching the reference (Fig. 2, green vs. black) — so the radiation physics (correlated-k CH_4 shortwave, mean-field haze optics, CIA + correlated-k gas-line longwave, convective adjustment, energy closure) is *correct*. The mean-field cross-section fixed the optical-depth *magnitude* (visible column $\tau \approx 8$, matching observation, versus ~ 70 before) and warmed the stratopause by ~ 30 K; the remaining gap is the haze *vertical distribution*, which is set by the Step 2 microphysics ($\rho_h \propto 1/\omega$) and differs from the observed profile. Closing it is a microphysics task (the haze production/settling that sets where the optical depth sits), not a radiation one. The validated radiation engine, energy closure, and haze→optics→DISORT coupling are in place, which is what Step 3 requires.

3.2 Cross-comparison with a reference radiation model

As an independent check we compare against a mature Fortran radiation model of the TAM lineage (Lombardo and Lora, 2023, the same scheme family as Lombardo & Lora 2023), a full correlated-k treatment (CH_4, C_2H_2, C_2H_6, C_2H_4, HCN plus N_2–N_2/N_2–CH_4/CH_4–CH_4/N_2–H_2 CIA) with prescribed haze, run to radiative–convective equilibrium on a 100-layer column under diurnally-averaged insolation (the diurnal cycle disabled for faster convergence). Both models are at steady state. We run our DISORT column on the *same* grid—100 layers spaced equally in $\log p$ from the 1 Pa model top to the surface—so the comparison is not confounded by vertical resolution.

Figure 2 overlays three profiles on a shared pressure coordinate. Driven by the Step 2 microphysics haze with mean-field optics (red), our model has the right structure and a ~ 140 K stratopause; driven instead by the *prescribed* haze (green)—the observational optical depths the reference uses, in the same radiative engine—it reaches ~ 200 K and tracks the reference (black) closely. This isolates the discrepancy cleanly: the radiation is right (the engine reproduces the reference given the same haze), so the residual red–green gap is the microphysics haze. The mean-field cross-section already corrected its *magnitude* (the column optical depth, previously $\sim 8\times$ too large); what remains is the haze *vertical distribution*—the microphysics deposits its

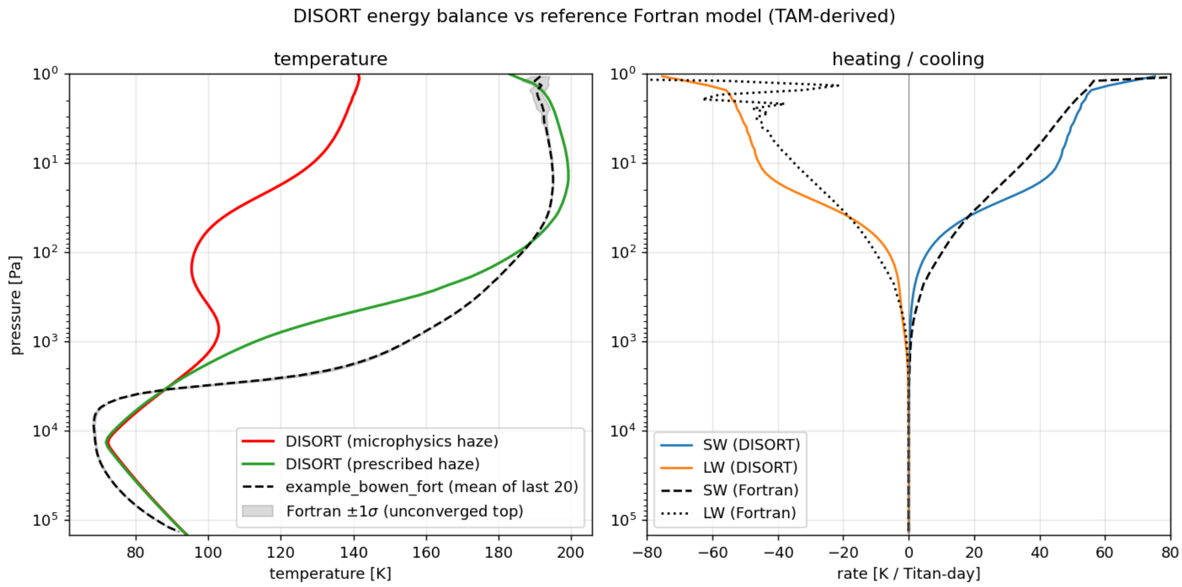

Figure 2: Our DISORT energy balance with the Step 2 *microphysics* haze (red) and with the observational *prescribed* haze (green), versus the reference TAM-lineage correlated-k Fortran model (black, time-mean of the last 20 snapshots), on a shared pressure axis. With the same haze the engines agree (green vs. black); the red–green gap is the microphysics haze. Right: shortwave heating and longwave cooling. The grey band is the reference's $\pm 1\sigma$ snapshot scatter: its explicit forward-Euler stepper formerly oscillated by up to ~ 24 K above ~ 4 Pa; a per-layer step cap reduces this to ~ 2.5 K at the model top with the converged profile unchanged, so the upper-atmosphere comparison is now against an essentially converged reference.

optical depth deeper than the observed profile—which is governed by the Step 2 haze production and settling, not by the radiation.

3.3 Per-source opacity breakdown

To localise the residual heating-rate differences we compare the two engines' optical depth *source by source* at the same state (the reference's converged $P(T)$ profile and the shared prescribed haze), accumulating each contribution—haze, CH_4 shortwave gas, IR gas lines ($CH_4/C_2H_2/C_2H_6/C_2H_4$/HCN), Rayleigh, and CIA—separately from the top down (Fig. 3). For this we enabled the reference model's per-band Gauss-weighted gas-opacity output, so even the correlated-k gas can be compared directly rather than inferred.

The prescribed haze, which both models read from the same observational tables, interpolates identically: at matched band centres the per-band optical depth agrees to machine precision (ratio 1.000) in both the shortwave and the longwave, at every level. The CH_4 correlated-k gas likewise agrees at the column scale (shortwave 1.03, longwave 1.08). The one source that genuinely differed is the collision-induced absorption (CIA), which lives in the deep atmosphere ($P \gtrsim 10$ kPa): there the breakdown showed our continuum running well above the reference's.

We traced this to the CIA formulation. Our model band-averaged the HITRAN cross-sections and formed $\tau = \sum_{\text{pairs}} k\, n_A n_B\, \Delta z$, whereas the reference evaluates a three-term exponential-sum fit to the band *transmission*, $\mathcal{T} = \sum_i a_i \exp(-k_i u)$ with $\tau = -\ln \mathcal{T}$ and absorber amount $u = n_1 n_2\, \Delta z$—a form that captures the sub-band absorber-amount nonlinearity a single band-averaged cross-section cannot. On the shared Titan far-IR continuum the band-averaged cross-section overestimated the optical depth by $\sim 1.75\times$. Adopting the reference's exponential-sum fits (reading the same transmission tables) brings the two continua into agreement to a few percent through the bulk of the deep atmosphere (Fig. 3, bottom row; the per-band cumulative-τ

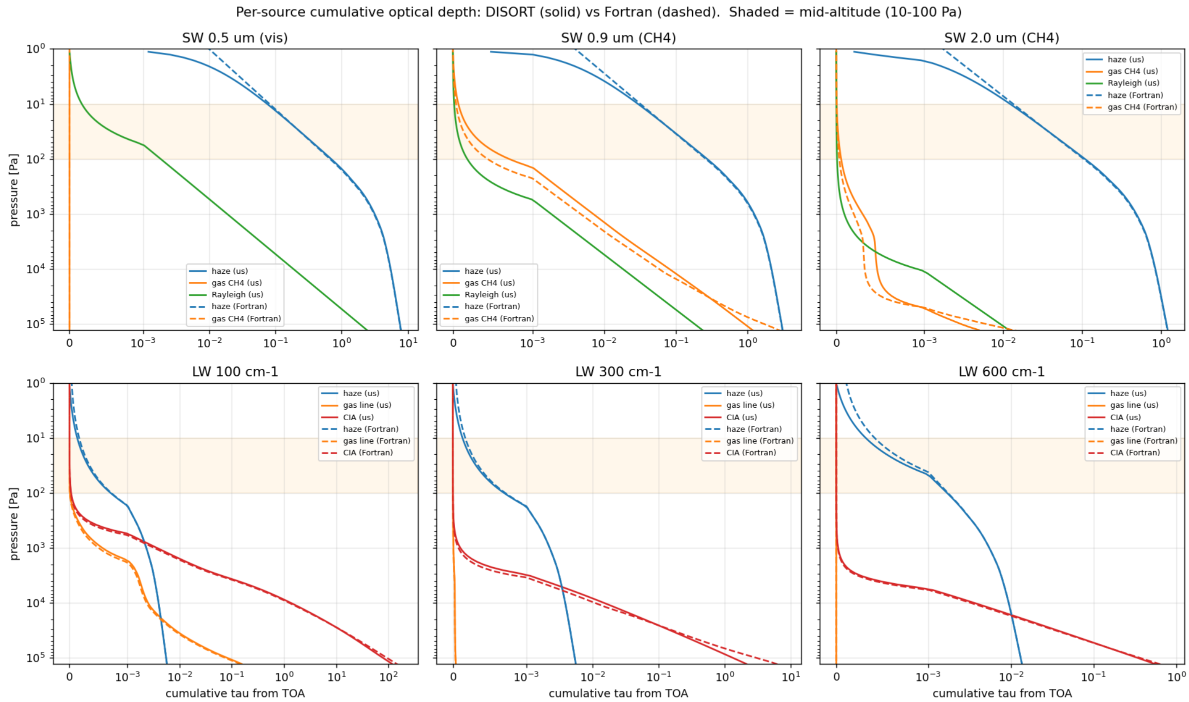

Figure 3: Cumulative optical depth from the top of atmosphere, by source, for our DISORT optics (solid) and the reference Fortran model (dashed), at the same state. Top row: shortwave (haze, CH_4 gas, Rayleigh) in three representative bands; bottom row: longwave (haze, gas lines, CIA). The shaded band marks the mid-altitude 10–100 Pa region. The haze dominates the mid-atmosphere and matches closely; the CIA continuum (bottom row, deep atmosphere) matches once our model adopts the reference's exponential-sum transmission fits in place of band-averaged HITRAN cross-sections.

ratio is 1.00 at 30 kPa, with the residual confined to the single lowest layer). With this, *every* opacity source—haze, CH_4, gas lines, Rayleigh, and CIA—matches the reference to within a few percent through the column, so the engines now differ only in the radiative-transfer solution itself, not in the optical depths fed to it.

4 Microphysical scaling law (Step 2)

We seek the steady population $C(z, N)$ of aggregates containing N monomers at altitude z (positive upward), built from production, Brownian coagulation, and gravitational sedimentation, with T the temperature, d the monomer radius, and D the fractal dimension. The physical ingredients—kernels, settling velocity, and fractal radius laws—follow Burgalat and Rannou (2017), with parameters from de Batz de Trenquelléon et al. (2025).

4.1 Monodisperse, self-similar closure

We adopt a monodisperse closure: at each altitude the aggregates share a single characteristic monomer count $\bar{N}(z)$,

$$C(z, N) = n(z)\,\delta\big(N - \bar{N}(z)\big)\,, \tag{1}$$

so the unknown fields are the number density $n(z)$ $[\mathrm{m}^{-3}]$ and the mean size $\bar{N}(z)$. The count $\bar{N}$ is treated as a *continuous monomer-volume count*: it measures particle volume in units of one monomer's volume, so $\bar{N} < 1$ denotes a sub-monomer sphere of radius $r = d\,\bar{N}^{1/3} < d$. This keeps mass bookkeeping exact through both coalescence (merging spheres, $\bar{N} < 1$) and aggregation (sticking, $\bar{N} \geq 1$). The closure is exact for the moments $M_0 = n$ and $M_1 = n\bar{N}$; it approximates the kernel by evaluating it at the mean size.

4.2 Fractal geometry and transport coefficients

An aggregate of N monomers of radius d and fractal dimension D has a mass (volume-equivalent) radius and a mobility (drag) radius

$$r(N) = d\,N^{1/3}, \qquad r_a(N) = d\,N^{1/D}, \qquad V(N) = \tfrac{4}{3}\pi d^3 N, \tag{2}$$

with the apparent–bulk conversion $r_a = E^{-1} r^{3/D}$, $E = d^{(D-3)/D}$ (Burgalat and Rannou, 2017). We carry two growth phases explicitly,

$$D(\bar{N}) = \begin{cases} 3 & \bar{N} < 1 \quad \text{(compact coalesced spheres, } E = 1,\ r_a = r\text{)}, \\ D & \bar{N} \geq 1 \quad \text{(fractal aggregates, e.g. } D = 2\text{)}, \end{cases} \tag{3}$$

the switch at $\bar{N} = 1$ being continuous in r_a (both give $r_a = d$).

The Stokes settling velocity with first-order Cunningham slip, generalized to fractal D (Burgalat and Rannou, 2017, Eq. 29), is

$$\omega(N, z) = \frac{2\rho_p\, g(z)\, d^2}{9\,\eta(z)} \left[N^{(D-1)/D} + \frac{A\,\lambda(z)}{d}\, N^{(D-2)/D} \right], \tag{4}$$

with $A = 1.591$, gas viscosity $\eta(z) = \eta(T)$, mean free path $\lambda(z) = \lambda(T, P)$, gravity $g(z)$, and $D = D(\bar{N})$. The equal-size Brownian coagulation kernel is bridged across the Knudsen regimes by the Fuchs harmonic mean of its continuum and free-molecular forms,

$$\beta_{\mathrm{CO}}(N, N) = \frac{2k_B T}{3\eta}\,(r_{ai} + r_{aj}) \left(\frac{1}{r_{ai}} + \frac{1}{r_{aj}} \right) \cdot [\text{slip}], \tag{5}$$

$$\beta_{\mathrm{FM}}(N, N) = \left(\frac{6k_B T}{\rho_p} \right)^{1/2} (r_{ai} + r_{aj})^2 \left(r_i^{-3} + r_j^{-3} \right)^{1/2}, \tag{6}$$

$$\beta(N, N; z) = \frac{\beta_{\mathrm{CO}}\,\beta_{\mathrm{FM}}}{\beta_{\mathrm{CO}} + \beta_{\mathrm{FM}}}. \tag{7}$$

Both ω and β are thus explicit functions of $(\bar{N}, z, T, d, D)$; the temperature enters through $k_B T/\eta$, $(k_B T/\rho_p)^{1/2}$, $\eta(T)$, and $\lambda(T, P)$.

4.3 Governing boundary-value problem

Working with the aggregate number density n and the monomer-volume density $M \equiv n\bar{N}$ (mass $= m_1 M$, $m_1 = \rho_p \frac{4}{3}\pi d^3$), the steady 1-D balance for any density q with downward settling speed ω and eddy diffusivity $K(z)$ reads $\mathrm{d}_z[\omega q + K\,\mathrm{d}_z q] = L_q - S_q$, where $\omega q + K\,\mathrm{d}_z q$ is the downward flux, S_q the production, and L_q the loss. Applied to the two densities,

$$\frac{\mathrm{d}}{\mathrm{d}z}\left[\omega(\bar{N}, z)\, M + K\, \frac{\mathrm{d}M}{\mathrm{d}z} \right] = -\, S_M(z), \tag{8}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\left[\omega(\bar{N}, z)\, n + K\, \frac{\mathrm{d}n}{\mathrm{d}z} \right] = \tfrac{1}{2}\,\beta(\bar{N}, z)\, n^2 - S_n(z). \tag{9}$$

Equation (8) expresses monomer conservation: coagulation merges particles but preserves total monomer volume, so the only monomer source is production. Equation (9) expresses Smoluchowski number loss—each collision removes one aggregate at rate $\frac{1}{2}\beta n^2$—plus seeding. With

Table 1: Asymptotic scaling exponents for equal-size aggregates, with depth $\equiv z_0 - z$. The spherical phase ($\bar{N} < 1$) uses $D = 3$.

Regime	$\beta(\bar{N}) \propto$	$\omega(\bar{N}) \propto$	$\mathrm{d}\bar{N}/\mathrm{d}z \propto -\bar{N}^s$, $s =$	$\bar{N}(\text{depth})$, $D = 2$
Continuum (Kn$\ll$1, low alt.)	$\bar{N}^0$	$\bar{N}^{(D-1)/D}$	$-2(D-1)/D$	$\bar{N} \propto \text{depth}^{1/2}$
Free-molecular (Kn$\gg$1, high alt.)	$\bar{N}^{2/D-1/2}$	$\bar{N}^{(D-2)/D}$	$(6-2D)/D - 1/2$	$\bar{N} \propto \text{depth}^2$

$\bar{N} = M/n$ and $D = D(\bar{N})$, this is a coupled two-point boundary-value problem for $M(z), n(z)$. The production source is a Gaussian (de Batz de Trenquelléon et al., 2025, Eq. 5),

$$S_M(z) = \frac{\mathcal{Q}_p}{m_1}\, G(z), \quad G(z) = \frac{1}{\sqrt{2\pi}\,\Delta z} \exp\left[-\frac{(z-z_0)^2}{2\,\Delta z^2}\right], \quad S_n = \frac{S_M}{N_{\text{seed}}}, \tag{10}$$

with $N_{\text{seed}} = (r_p/d)^3$. Boundary conditions are vanishing densities at the model top and settling-only deposition ($\mathrm{d}_z M = \mathrm{d}_z n = 0$) at the surface. A first integral of (8) well below the source recovers the constant downward monomer flux $\omega M + K\, \mathrm{d}_z M = P$, with $P = \mathcal{Q}_p/m_1$.

4.4 Sedimentation-dominated limit: the master scaling ODE

Setting $K \to 0$ yields a closed-form law—both a physical limit and the initial guess for the BVP solver. Monomer-flux constancy gives the classical result that the haze mass density equals the mass flux divided by the fall speed,

$$\omega\, n\, \bar{N} = P \implies n(z) = \frac{P}{\omega\, \bar{N}}, \qquad \rho_h(z) = \frac{\mathcal{Q}_p}{\omega(\bar{N}, z)}. \tag{11}$$

Substituting $\omega n = P/\bar{N}$ into (9) (below the source) collapses the system to a single first-order ODE for the mean size,

$$\boxed{\frac{\mathrm{d}\bar{N}}{\mathrm{d}z} = -\frac{\beta(\bar{N}, z)\, P}{2\,\omega(\bar{N}, z)^2}}, \qquad \bar{N}(z_0) = N_{\text{seed}}, \tag{12}$$

integrated downward and switching $D(\bar{N})$ from 3 to D at $\bar{N} = 1$. Because the right-hand side is negative, $\bar{N}$ grows as the particle falls; the number density and sizes then follow from (11) and (2).

4.5 Asymptotic scalings

Holding the background quantities $(T, \eta, \lambda, g, \rho_p, P)$ roughly constant over a scale height, β and ω become pure powers of $\bar{N}$ and (12) integrates to power laws (Table 1). Just below the production region the haze is free-molecular and the mean size grows as $\bar{N} \propto (\text{depth})^2$; deep in the dense lower atmosphere the continuum kernel is size-independent and growth slows to $\bar{N} \propto (\text{depth})^{1/2}$. The full Fuchs kernel (7) interpolates smoothly between these limits, which is why (12)/(8)–(9) are integrated numerically rather than reduced to a single exponent.

4.6 Numerical solution

We integrate the master ODE (12) downward from $z_0 = 415\,\text{km}$ (seed $N_{\text{seed}} = (r_p/d)^3 \approx 8 \times 10^{-3}$) on the crude Titan reference column, with $\eta(T)$ from a Sutherland law and $\lambda(T, P)$ from kinetic theory (Table 2). Figure 4 shows the result. The mean size grows by seven orders of magnitude from the sub-monomer seed to $\bar{N} \sim 7 \times 10^4$ at the surface; the spherical→fractal switch at $\bar{N} = 1$ near 390 km is visible as the divergence of the mobility radius r_a from the

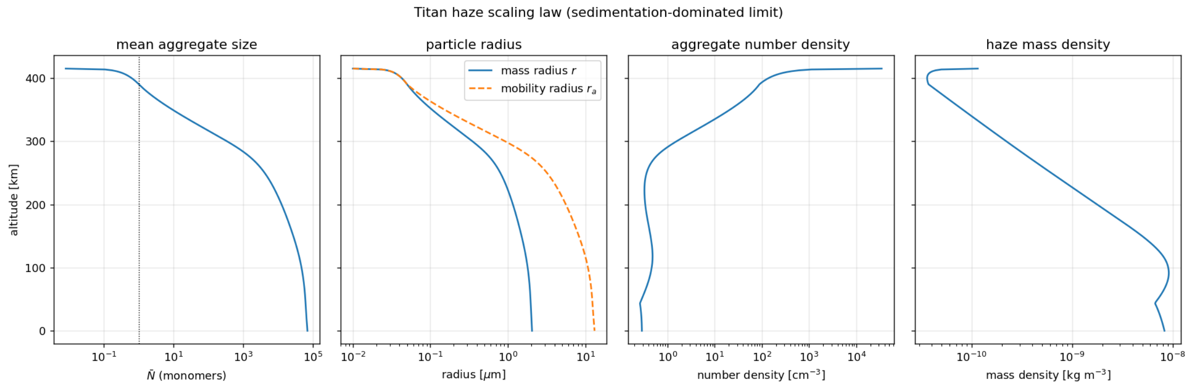

Figure 4: Steady haze profiles from the sedimentation-dominated scaling law (12) on the Titan reference column: mean aggregate size $\bar{N}$ (dotted line marks the $\bar{N} = 1$ spherical/fractal switch), mass and mobility radii, aggregate number density, and haze mass density. Aggregates grow as they settle; r_a exceeds r in the fractal phase.

mass radius r (open aggregates have $r_a > r$ for $D < 3$). The mass radius reaches $\approx 0.35\,\mu$m near 300 km—consistent with the characteristic radius $r_c \approx 0.46\,\mu$m of de Batz de Trenquelléon et al. (2025)—and $\approx 2\,\mu$m at the surface. Number densities span ~ 0.3–$3 \times 10^4\,\mathrm{cm}^{-3}$, and the monomer mass flux $\rho_h\,\omega$ is conserved to machine precision, confirming Eq. (11).

4.7 Eddy-diffusion BVP and cross-validation

Restoring eddy diffusion, we solve the full boundary-value problem (8)–(9) on $[0, z_0]$, imposing the (narrow) production as a top flux boundary condition—downward monomer flux $\Phi_M = P$ and seed number flux $\Phi_n = P/N_{\mathrm{seed}}$—and settling-only deposition at the surface. The $K \to 0$ master ODE supplies the initial guess. A Titan-like eddy profile $K(z) = K_0 \exp(z/H_K)$ with $K_0 = 1\,\mathrm{m^2s^{-1}}$ and $H_K = 150$ km keeps K small ($\lesssim 16\,\mathrm{m^2s^{-1}}$) across the haze, as expected in Titan's stratosphere.

We cross-validate the resulting profile against two published constraints (Fig. 5). First, the haze extinction scale height: fitting $n\,r_a^2$ over 80–300 km yields $H = 64$ km, against the 65 km inferred by Tomasko et al. (2008); the $K \to 0$ settling limit alone gives 54 km, so the modest eddy diffusion brings the model into close agreement. Second, the characteristic size: the mass radius is $\approx 0.34\,\mu$m near 300 km and sub-micron through the main haze, consistent with the $r_c \approx 0.46\,\mu$m of de Batz de Trenquelléon et al. (2025). The monomer mass flux is conserved to machine precision (global mass balance: surface deposition equals column production), and the BVP departs from the master ODE by only $\sim 9\%$ in the settling-dominated lower haze. The extinction scale height is sensitive to $K(z)$, identifying the eddy profile as the principal uncertain input.

5 Coupled iteration (Step 3)

Per layer the microphysics supplies the number density $n(z)$ and the characteristic radius $r_a(z) = d\,\bar{N}^{1/D}$. These map to layer optical properties via a precomputed aggregate-optics table (de Batz de Trenquelléon et al., 2025, Eqs. 16–18): the optical thickness is $\Delta\tau(\lambda, z) = n(z)\,C_{\mathrm{ext}}(r_a, \lambda)\,\Delta z$, with mean single-scattering albedo and asymmetry parameter obtained by opacity weighting. The aggregate projected area scales as $\bar{N}^{2/D}d^2$. The feedback loop is then

$$T(z) \longrightarrow \eta(T), \lambda(T,P) \xrightarrow{(12)/\S 4} n(z), \bar{N}(z) \xrightarrow{\text{optics}} \Delta\tau, \omega_0, g \xrightarrow{\text{RT engine}} T(z),$$

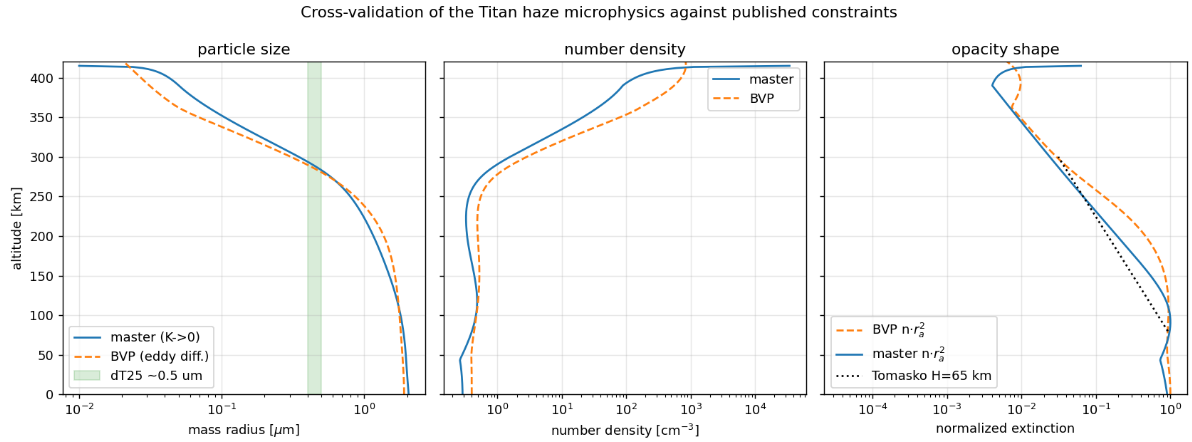

Figure 5: Cross-validation of the microphysics against published Titan constraints. Left: mass radius from the $K \to 0$ master ODE and the eddy-diffusion BVP, with the de Trenquelléon et al. (2025) main-haze size ($\sim 0.5\,\mu$m) shaded. Centre: number density. Right: normalized extinction $n\,r_a^2$, with the Tomasko et al. (2008) $H = 65\,$km reference slope (dotted) anchored at 80 km.

iterated under-relaxed. The sign and strength of the feedback are read from the iteration sequence—which, as we show, does not converge to a single fixed point but exposes a bistable system.

5.1 Implementation and the coupled iteration

We drive the Step 1 engine with the Step 2 haze through its prescribed-haze input: each pass maps the microphysics to the engine's per-band, per-level optical-depth tables (the same mean-field RDG cross-section used in §3, with the single-scattering albedo and asymmetry kept at the observational spectral shape; the longwave is a pure absorber), re-runs the model to radiative–convective equilibrium, and feeds the new $T(z)$ back to the microphysics, under-relaxed. The interface is validated end-to-end: writing the *observational* haze through the same path reproduces the stock prescribed $T(z)$ to 0.000 K, and the microphysics-haze tables reproduce the optical depth of the in-line DISORT coupled run to machine precision. The engine top is first stabilised (a per-layer time-step cap reduces its $\lesssim 4\,$Pa oscillation from ~ 24 to $\sim 2.5\,$K), since the haze source sits there.

5.2 Result: a bistable haze feedback

The coupled iteration does *not* settle to a single fixed point. It oscillates between two states (Fig. 6): a *warm* branch with the stratopause near 183 K at 8 Pa (close to the prescribed-haze equilibrium) and a *cool* branch near 131–135 K at 1.3–1.7 Pa (the microphysics-haze state). The branches are anti-correlated under naive iteration, and the transition between them is sharp: near $T \approx 137\,$K a $\sim 1\,$K change in the input profile flips the equilibrium output by $\sim 48\,$K.

5.3 Diagnosis: a genuine absorbing-haze radiative bistability

The composite map is $T \to \text{haze} \to T$; we isolate each half to locate the sharp transition. *First*, the microphysics is smooth: sweeping input temperature profiles across stratopause 128–225 K and recomputing the haze for each, the column optical depth, the τ-weighted centroid pressure, and the haze mass column all vary continuously — largest adjacent change $\leq 3.5\%$, with no feature at 137 K (Fig. 7). The haze centroid *rises* smoothly with temperature ($1714 \to 770\,$Pa

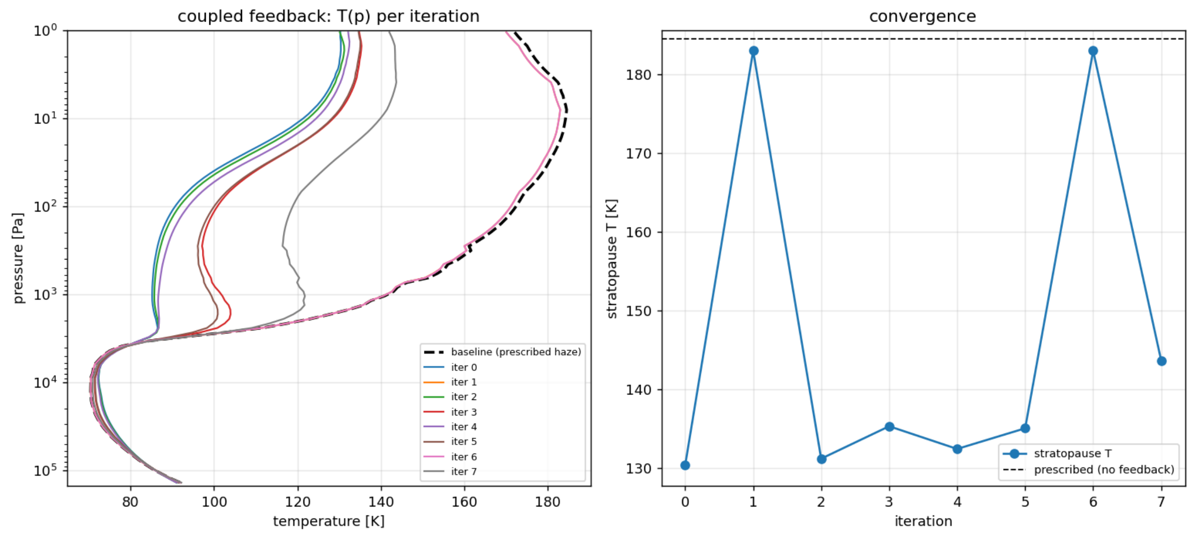

Figure 6: Coupled radiative–microphysical iteration. Left: temperature profile per iteration (dashed: the prescribed-haze, no-feedback baseline). Right: the stratopause temperature does not converge but alternates between a warm ($\sim$183 K) and a cool ($\sim$131 K) branch.

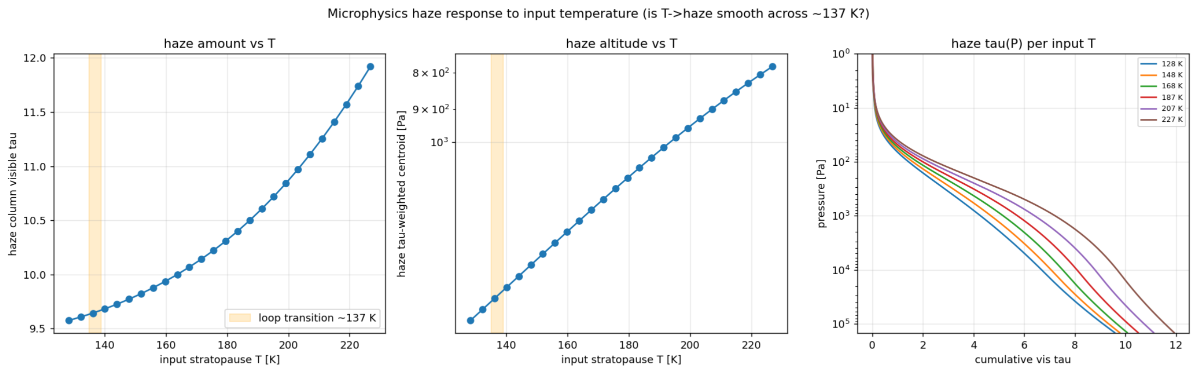

Figure 7: Microphysics haze response to the input temperature. Haze column optical depth (left), τ-weighted centroid pressure (centre), and cumulative $\tau(p)$ (right) all vary smoothly across the coupled transition (shaded, $\sim$137 K): the $T \to$haze map has no cliff, so the bistability is radiative, not microphysical.

over that range), the signature of a positive feedback: a warmer stratosphere lifts the absorbing haze, which then absorbs higher and warms it further. *Second*, the radiative transfer is genuinely multi-valued: holding the haze *fixed* (the transition-state haze) and relaxing the radiative–convective equilibrium from a warm versus a cool initial state lands on two distinct equilibria, stratopause 158 K versus 141 K ($\Delta T \approx 31$ K); the separation is stable as the residual falls, so it is real multiplicity rather than under-convergence, and it reproduces in the independent DISORT engine.

Together these identify the 137 K transition as a *genuine radiative feedback*, not a microphysics regime change or a coupling artifact: the strongly-absorbing haze, smoothly mobile in altitude with temperature, gives the one-dimensional radiative–convective system two stable states with gain > 1 between them — the absorbing-aerosol multiple-equilibria mechanism, here driven self-consistently by the aerosol microphysics. A single converged branch (and the feedback gain along it) therefore requires a continuation or damped-Newton solve of $T = F(\text{haze}(T))$ that tracks one attractor rather than fixed-point iteration; the existence of the two branches is

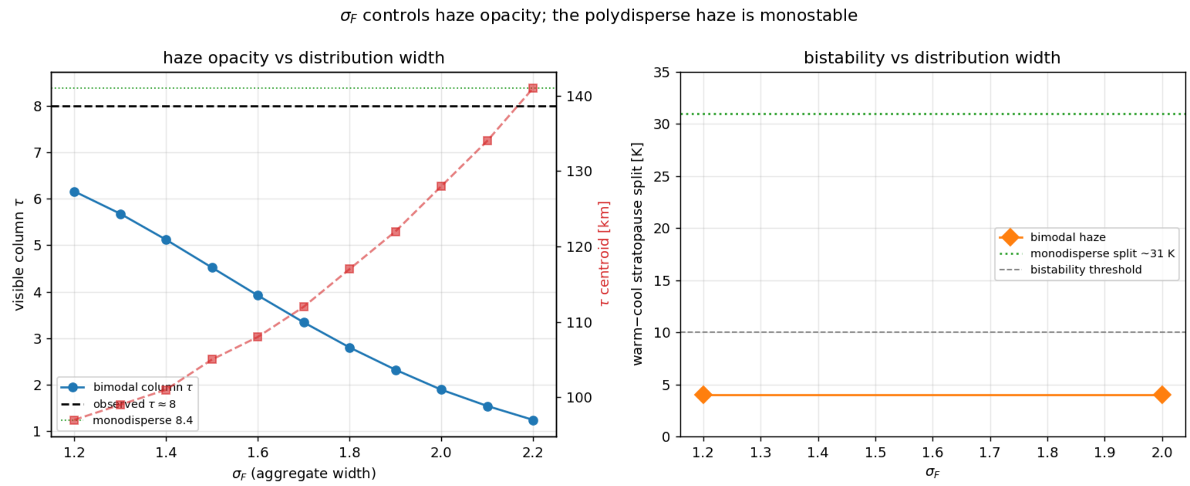

Figure 8: Bimodal (polydisperse) haze versus the aggregate geometric width σ_F. Left: the visible column optical depth falls steeply with σ_F (gravitational sorting), reaching the observed $\tau \approx 8$ only for a narrow distribution $\sigma_F \lesssim 1.2$; the τ-weighted centroid (dashed, right axis) rises as the haze thins. Right: the warm−cool stratopause split (the bistability indicator) stays near 4 K across σ_F—monostable—versus the $\sim$31 K split of the monodisperse haze. The absorbing-haze bistability is an artifact of the single-size closure, not a robust feature of the coupled system.

itself the principal Step-3 result.

5.4 Polydispersity: the bistability does not survive

The diagnosis above assumes the monodisperse closure of §4: one size $\bar{N}(z)$ per level. Because the bistability is an opacity-driven feedback, it is worth asking whether a realistic *size distribution* changes the picture. We replaced the closure with a bimodal two-moment log-normal scheme (a spherical-monomer mode and a fractal-aggregate mode, each carried by its number and volume moments, with Brownian coagulation, an inter-mode coalescence transfer, and size-dependent settling), following Burgalat and Rannou (2017). Two effects appear that the single-size model cannot represent. First, *gravitational sorting*: the moment-averaged settling velocities differ, $\langle\omega\rangle_3 > \langle\omega\rangle_0$, so the broad aggregate distribution sheds its mass faster than a single particle of the mean size and less haze accumulates aloft. The visible column optical depth then falls steeply with the geometric width σ_F (Fig. 8, left): from $\tau \approx 8.4$ in the monodisperse limit to 6.2 at $\sigma_F = 1.2$ and 1.9 at $\sigma_F = 2.0$. Matching the observed $\tau \approx 8$ requires a narrow distribution, $\sigma_F \lesssim 1.2$, so the haze opacity becomes a *constraint* on the aggregate size spread.

Second, and decisively, the *bistability does not persist*. Repeating the fixed-haze warm-versus-cool equilibrium test of §5.3 with the bimodal haze gives a single equilibrium at every width sampled—the warm and cool starts converge to within $\sim$4 K at both $\sigma_F = 1.2$ (the thick, nearly-observed haze) and $\sigma_F = 2.0$ (Fig. 8, right), versus the $\sim$31 K split of the monodisperse haze. The polydisperse haze, even at the larger optical depth of the narrow-distribution limit, stays below the feedback threshold. The two-equilibria behaviour is therefore a feature of the monodisperse idealization—its over-concentrated opacity at altitude—rather than a robust property of Titan's coupled haze, a caution against reading strong nonlinear feedbacks out of single-size microphysics.

Table 2: Baseline microphysical and atmospheric parameters adopted for Titan. Sources: T08 (Tomasko et al., 2008), L23 (Lombardo and Lora, 2023), BR17 (Burgalat and Rannou, 2017), dT25 (de Batz de Trenquelléon et al., 2025).

Quantity	Symbol	Value	Source
Fractal dimension	D	2.0 (free, 2–3; 3 $\Rightarrow$ spheres)	dT25
Monomer radius	d	50 nm	dT25 (T08)
Production seed radius	r_p	10 nm	dT25
Aerosol density	ρ_p	$800\,\mathrm{kg\,m^{-3}}$	dT25
Mass production rate	$\mathcal{Q}_p$	$2.1 \times 10^{-13}\,\mathrm{kg\,m^{-2}\,s^{-1}}$	dT25 (T08)
Production altitude	z_0	$415\,\mathrm{km} \approx 1\,\mathrm{Pa}$	dT25
Production width	Δz	20 km	dT25
Charge density	n_e	$15\,e^-\,\mu\mathrm{m}^{-1}$	dT25
Slip-flow constant	A	1.591	BR17
Surface temperature	T_s	93.5 K	T08
Surface pressure	P_s	$\sim$1.47 bar	L23
Stratospheric CH_4	–	1.4%	T08
Haze opacity scale height	H	65 km (above 80 km)	T08
RT grid	–	0–540 km, $\sim$50 layers	T08, L23
Shortwave / longwave split	–	$5\,\mu$m	L23

6 Photochemistry in the loop (Step 4)

In Steps 1–3 the production rate $\mathcal{Q}_p$ and seed properties are imposed ($\mathcal{Q}_p = 2.1\times10^{-13}\,\mathrm{kg\,m^{-2}\,s^{-1}}$ at $z_0 \approx 415\,\mathrm{km}$; de Batz de Trenquelléon et al., 2025). Step 4 promotes $\mathcal{Q}_p$ to an output of a photochemical module driven by CH_4/N_2 photolysis, so the haze mass flux responds to the radiation field and composition, completing the radiation$\leftrightarrow$chemistry$\leftrightarrow$microphysics feedback.

7 Baseline parameters

Status and approximations. The monodisperse closure replaces the true size spread by $\bar{N}$ and evaluates the kernel at the mean size; real distributions coagulate somewhat faster, an $O(1)$ correction absorbable into a prefactor on β and calibratable against a full population-balance run. The spherical$\rightarrow$fractal switch at $\bar{N} = 1$ is treated as instantaneous. Background profiles $\eta(T)$, $\lambda(T, P)$ are taken from N_2 kinetic theory, $g(z)$ from Titan gravity, and the eddy diffusivity $K(z)$ from a standard Titan profile (Vuitton et al., 2019). Charge inhibition (a multiplicative factor on β) is deferred. With Steps 1–3 implemented and the bistable feedback diagnosed, the immediate next steps are a continuation/damped-Newton solve to trace a single branch and quantify the feedback gain along it, and the Step 4 photochemical production (Sec. 6); charge inhibition and a full population balance are the main physics refinements thereafter.

References

J. Burgalat and P. Rannou. Brownian coagulation of a bi-modal distribution of both spherical and fractal aerosols. *Journal of Aerosol Science*, 105:151–165, 2017. doi: 10.1016/j.jaerosci.2016.12.001.

M. Cabane, P. Rannou, E. Chassefière, and G. Israel. Fractal aggregates in Titan's atmosphere. *Planetary and Space Science*, 41(4):257–267, 1993. doi: 10.1016/0032-0633(93)90021-S.

B. de Batz de Trenquelléon, P. Rannou, S. Vinatier, et al. The new Titan planetary climate model. II. Titan's haze and cloud cycles. *The Planetary Science Journal*, 6(4):79, 2025. doi: 10.3847/PSJ/adbb6c.

T. Karman, I. E. Gordon, A. van der Avoird, et al. Update of the HITRAN collision-induced absorption section. *Icarus*, 328:160–175, 2019. doi: 10.1016/j.icarus.2019.02.034.

N. A. Lombardo and J. M. Lora. Influence of seasonally varying atmospheric composition on the radiative energy budget of Titan's stratosphere. *Icarus*, 390:115291, 2023. doi: 10.1016/j.icarus.2022.115291.

P. Rannou, F. Hourdin, C. P. McKay, and D. Luz. A coupled dynamics-microphysics model of Titan's atmosphere. *Icarus*, 170(2):443–462, 2004. doi: 10.1016/j.icarus.2004.03.007.

K. Stamnes, S.-C. Tsay, W. Wiscombe, and K. Jayaweera. Numerically stable algorithm for discrete-ordinate-method radiative transfer in multiple scattering and emitting layered media. *Applied Optics*, 27(12):2502–2509, 1988. doi: 10.1364/AO.27.002502.

M. G. Tomasko, B. Bézard, L. Doose, S. Engel, E. Karkoschka, and S. Vinatier. Heat balance in Titan's atmosphere. *Planetary and Space Science*, 56(5):648–659, 2008. doi: 10.1016/j.pss.2007.10.012.

V. Vuitton, R. V. Yelle, S. J. Klippenstein, S. M. Hörst, and P. Lavvas. Simulating the density of organic species in the atmosphere of Titan with a coupled ion-neutral photochemical model. *Icarus*, 324:120–197, 2019. doi: 10.1016/j.icarus.2018.06.013.